School Management System - Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a comprehensive School Management System designed for educational institutions in Bangladesh.

1.2 Scope

The system will digitize and automate school operations including admission, attendance, fee management, academic tracking, communication, and administrative tasks.

1.3 Target Users

- School Administrators
- Teachers
- Students
- Parents/Guardians
- Accountants/Finance Staff
- Office Staff

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2. Feature Prioritization & Implementation Phases

PHASE 1: Core/MVP Features (Must-Have)

Essential features for basic school operations

FR-1.1: User Management & Authentication

- Multi-role user system (Admin, Teacher, Student, Parent, Staff)
- Secure login with role-based access control (RBAC)
- User profile management
- Password reset functionality

FR-1.2: Student Information Management

- Student registration and profile creation
- Student personal information storage
- Class/section assignment

- Student ID card generation (#13)
- Document storage (birth certificate, certificates, photos) (#24)

FR-1.3: Admission Management (#1)

- Online admission application form
- Offline admission form download (PDF)
- Admission application review and approval workflow
- Admission fee collection
- Student enrollment confirmation

FR-1.4: Attendance Management (#2)

- Digital attendance for students (class-wise)
- Digital attendance for teachers
- Daily/monthly attendance reports
- Attendance analytics and summary

FR-1.5: Fee Management (#8, #9)

- Fee structure configuration (tuition, exam, monthly fees)
- Fee collection and receipt generation
- Due/outstanding balance tracking
- Payment history
- Fee category management (class tests, tours, etc.)

FR-1.6: Class & Section Management

- Class creation and management
- Section/division management
- Class strength tracking
- Student-class assignment

FR-1.7: Academic Year Management

- Academic session/year setup
- Year transition and rollover

FR-1.8: Basic Reporting

- Student list reports

- Attendance reports
- Fee collection reports
- Dues/outstanding reports (#9)

PHASE 2: Important Features (High Priority)

Features that significantly enhance functionality

FR-2.1: Exam & Result Management

- Exam schedule creation
- Mark entry system
- Result card generation
- Grade/GPA calculation
- Progress reports

FR-2.2: Class Routine Management (#7)

- Class timetable creation
- Teacher assignment to periods
- Room allocation
- Routine printing and download
- Routine display for students/teachers

FR-2.3: Leave Management (#3)

- Leave application submission (students & teachers)
- Leave approval workflow
- Leave balance tracking
- Leave history and reports

FR-2.4: Notification System (#4, #16)

- SMS notification integration
- Email notification system
- Absence notifications to parents
- Fee reminder notifications
- General announcements

- Category-wise notices (teacher, student, staff)

FR-2.5: Homework & Study Tracking (#5)

- Homework assignment by teachers
- Homework submission tracking
- Daily progress logging
- Study status reports
- Parent dashboard with student progress

FR-2.6: Syllabus Management (#6)

- Curriculum/syllabus upload
- Today's lesson display
- Upcoming lessons/topics preview
- Exam schedule visibility
- Teacher preparation notes

FR-2.7: Certificate Generation (#11)

- Testimonial generation
- Transfer certificate (TC)
- Character certificate
- Course completion certificate
- Custom certificate templates

FR-2.8: Billing & Voucher Management (#10)

- Billing voucher generation
- Voucher approval workflow
- Bill payment tracking
- Invoice printing

FR-2.9: Teacher Management

- Teacher profile and credentials
- Subject assignment
- Schedule/timetable view
- Performance tracking (#28)

FR-2.10: Auto Promotion System (#12)

- Year-end automatic class promotion
- Promotion criteria configuration
- Board exam year handling
- Manual promotion override
- Promotion report generation

PHASE 3: Advanced Features (Medium Priority)

Features for comprehensive management

FR-3.1: Financial Management (#15)

- Expense recording and categorization
- Income tracking
- Profit/loss calculation
- Financial summary reports
- Budget management

FR-3.2: Salary Management (#26)

- Salary structure configuration
- Monthly salary processing
- Salary slip generation
- Salary payment tracking
- Deduction management

FR-3.3: Inventory Management (#14)

- Item distribution tracking (notebooks, uniforms, books, ID cards)
- Stock management
- Item-wise pricing
- Distribution reports
- Payment collection for items

FR-3.4: Fee Enforcement & Penalties (#27)

- Automatic fee overdue detection

- Auto-notice generation for overdue fees
- Admission suspension for non-payment
- Fine/penalty calculation
- Reinstatement with penalty payment

FR-3.5: Performance & Analytics (#28)

- Student performance reports
- Teacher performance evaluation
- Comparative analysis
- Trend reports
- Dashboard with key metrics

FR-3.6: Communication Hub

- Internal messaging system
- Teacher-parent communication
- Student-teacher messaging
- Group messaging
- Message history and archiving

FR-3.7: Meeting Management (#22)

- Meeting scheduling
- Meeting minutes recording
- Attendee tracking
- Meeting document storage
- Minutes printing/download

FR-3.8: Event Management (#30)

- Event planning and scheduling
- Role assignment for events
- Task distribution
- Event invitation generation (#23)
- Event invitation printing/download
- Event attendance tracking

FR-3.9: Notice Board System (#18)

- General notice creation and publishing
- Target-specific notices (individual teacher/student)
- Notice display on website/app
- Email notification for notices
- Notice archive

FR-3.10: School Timing Management (#32)

- Session timing configuration (winter/summer)
- Automatic routine time adjustment
- Multiple timing templates
- Schedule-based timing change

PHASE 4: Optional/Advanced Features (Nice-to-Have)

Features for enhanced experience

FR-4.1: Library Management

- Book catalog management
- Book issue/return tracking
- Fine calculation for late returns
- Library membership
- Book reservation system

FR-4.2: Transport Management

- Bus route management
- Student transport assignment
- Transport fee calculation
- Driver and vehicle tracking
- GPS tracking integration (future)

FR-4.3: Hostel Management (if applicable)

- Room allocation
- Hostel fee management

- Mess management
- Hostel attendance

FR-4.4: Canteen Management (#29)

- Menu management
- Order placement
- Payment tracking
- Inventory for canteen
- Meal plan management

FR-4.5: Scout/Girls Guide Management (#31)

- Member registration
- Activity tracking
- Badge/achievement recording
- Event participation
- Training records

FR-4.6: Managing Committee System (#25)

- Committee member management
- Meeting scheduling
- Decision tracking
- Document management
- Role and responsibility assignment

FR-4.7: Duty/Shift Management (#17)

- Shift roster creation
- Duty assignment (security guards, staff)
- Duty schedule display
- Duty completion tracking
- Shift swapping

FR-4.8: Online Learning Platform (#20)

- Live class scheduling and links
- Recorded class upload to YouTube

- Video library management
- Class recording archive
- Student video access tracking

FR-4.9: Email Integration (#19)

- School email system integration
- Email viewing within system
- Reply functionality
- Email categorization
- Official correspondence tracking

FR-4.10: Parent Portal & Support (#21)

- Dedicated parent dashboard
- Real-time student progress view
- Fee payment online
- Communication with teachers
- Support ticket system (email/form/live chat)
- Complaint and suggestion management

FR-4.11: Mobile Application

- Native iOS/Android apps
- Push notifications
- Offline data access
- QR code attendance
- Parent-teacher communication

FR-4.12: Document Management System

- Centralized document repository
- Version control for documents
- Document approval workflow
- Document search and filter
- Document sharing

FR-4.13: Online Examination System

- Question bank management
- Online test creation
- Auto-grading for MCQs
- Proctoring features
- Result publication

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3. Additional Recommended Features

FR-A1: Dashboard & Analytics

- Role-based customizable dashboards
- Real-time statistics
- Visual charts and graphs
- Quick action shortcuts
- Key performance indicators (KPIs)

FR-A2: Backup & Data Security

- Automated daily backups
- Data encryption
- User activity logs
- Audit trails
- GDPR/data privacy compliance

FR-A3: Multi-Language Support

- Bengali as primary language
- English as secondary
- Language switching option
- RTL support if needed

FR-A4: Report Builder

- Custom report creation
- Report scheduling
- Export to PDF/Excel
- Report templates

- Automated report generation

FR-A5: API & Integration

- RESTful API for third-party integration
- SMS gateway integration (local providers)
- Payment gateway integration (bKash, Nagad, bank)
- Government database integration (if applicable)
- Google Calendar/Workspace integration

FR-A6: Alumni Management

- Alumni registration
- Alumni database
- Alumni events
- Fundraising campaigns
- Career tracking

FR-A7: Grievance Redressal System

- Online complaint submission
- Complaint tracking and resolution
- Escalation workflow
- Feedback collection
- Resolution time tracking

FR-A8: Biometric Integration

- Fingerprint attendance device integration
- Face recognition (future)
- RFID card attendance
- Real-time sync with system

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4. Non-Functional Requirements

NFR-1: Performance

- Page load time < 3 seconds
- Support 500+ concurrent users

- Database query optimization
- Caching strategy implementation

NFR-2: Security

- HTTPS/SSL encryption
- SQL injection prevention
- XSS attack prevention
- CSRF token implementation
- Regular security audits
- Data backup and recovery plan

NFR-3: Scalability

- Horizontal scaling capability
- Database sharding support (for multi-tenant)
- Load balancing
- CDN for static assets

NFR-4: Usability

- Intuitive user interface
- Responsive design (mobile-friendly)
- Accessibility standards (WCAG 2.1)
- Minimal training required
- Consistent UI/UX across modules

NFR-5: Reliability

- 99.5% uptime
- Automated failover
- Error logging and monitoring
- Graceful error handling

NFR-6: Maintainability

- Modular architecture
- Code documentation
- Version control

- Automated testing (unit, integration)
- CI/CD pipeline

NFR-7: Compatibility

- Browser support (Chrome, Firefox, Safari, Edge)
- Mobile browser compatibility
- Cross-platform database support
- API backward compatibility

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5. System Architecture Recommendations

5.1 Backend Stack Options

Option 1: Laravel (Recommended for your skills)

- PHP 8.2+
- Laravel 10.x/11.x
- MySQL/PostgreSQL
- Redis for caching
- Queue system for background jobs

Option 2: Python (For learning)

- Django or FastAPI
- PostgreSQL
- Celery for async tasks
- Redis

5.2 Frontend

- Vue.js or React for SPA
- Blade templates for server-side rendering (Laravel)
- Tailwind CSS or Bootstrap 5
- Alpine.js for lightweight interactions

5.3 Mobile App

- React Native or Flutter
- API-first approach

5.4 Infrastructure

- Docker for containerization
- Nginx as reverse proxy
- Cloud hosting (AWS, DigitalOcean, or local providers)

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6. Database Design Considerations

Key Entities

- Users (with polymorphic relationships)
- Students
- Teachers
- Parents
- Classes/Sections
- Academic Sessions
- Attendance Records
- Fee Structures & Transactions
- Exams & Results
- Notices & Announcements
- Documents
- Audit Logs

Important Relationships

- Many-to-Many: Students ↔ Classes (for multiple subjects)
- One-to-Many: Parents ↔ Students
- One-to-Many: Teachers ↔ Classes
- Polymorphic: Documents ↔ (Students/Teachers/Staff)

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7. Development Approach Recommendation

Phase-wise Implementation

1. **Months 1-2**: Phase 1 (Core MVP)
2. **Months 3-4**: Phase 2 (High Priority Features)
3. **Months 5-6**: Phase 3 (Advanced Features)

4. **Months 7+**: Phase 4 (Optional Features) + Refinement

Agile Methodology

- 2-week sprints
- Regular demos with stakeholders
- Iterative feedback incorporation
- Test-driven development (TDD)

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8. Testing Requirements

- Unit testing (PHPUnit for Laravel)
- Integration testing
- User acceptance testing (UAT)
- Performance testing
- Security testing
- Browser compatibility testing

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9. Documentation Requirements

- User manuals (role-wise)
- Administrator guide
- API documentation
- Database schema documentation
- Deployment guide
- Troubleshooting guide

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10. Success Metrics

- User adoption rate
- System uptime
- Transaction success rate
- User satisfaction score
- Support ticket volume
- Feature utilization rate

Notes

This SRS prioritizes features based on:

1. **Criticality**: Essential for day-to-day operations
2. **Dependencies**: Features that other features depend on
3. **User Impact**: Maximum benefit to maximum users
4. **Implementation Complexity**: Easier features in earlier phases

The document is designed to be flexible and can be adapted based on specific school requirements and feedback.

Version: 1.0
Last Updated: November 6, 2024
Status: Draft